

Smart Temperature-Controlled Fan Using Arduino

Submitted By

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Objective

The objective of this project is to design and simulate a Smart Temperature-Controlled Fan using Arduino. The system monitors temperature using a TMP36 temperature sensor and automatically controls the fan speed according to the measured temperature. A status LED is used to indicate system operation.

Components Used

- Arduino Uno R3
- TMP36 Temperature Sensor
- L293D Motor Driver IC
- DC Motor (Fan)
- Green LED
- 220Ω Resistor
- Breadboard
- Jumper Wires
- 9V Battery

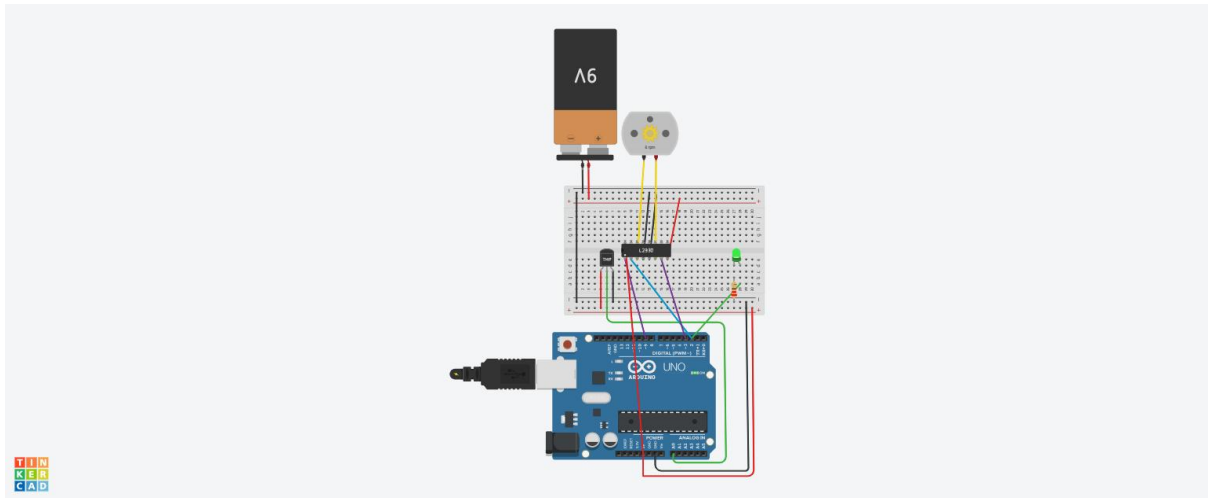
Circuit Description

The TMP36 temperature sensor is connected to the Arduino analog input pin A0. The Arduino continuously reads the sensor output voltage and converts it into temperature in degrees Celsius.

The DC motor is controlled through an L293D motor driver IC. PWM (Pulse Width Modulation) is applied to the enable pin of the motor driver to vary the fan speed based on temperature.

A green LED is used as a status indicator to show that the system is active and functioning properly.

Circuit Screenshot



Tinkercad Link

<https://www.tinkercad.com/things/4YS2bE0bvzC-grand-wluff-juttuli?sharecode=EitCoDIor9EepjuPRuLGt7m7kVjcxhPifhE-MqXaOrg>

Arduino Source Code

```
const int tempPin = A0;
```

```
const int motorEnable = 9;
```

```
const int motorIn1 = 2;
```

```
const int motorIn2 = 3;
```

```
const int greenLED = 8;
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  pinMode(motorEnable, OUTPUT);
```

```
  pinMode(motorIn1, OUTPUT);
```

```
  pinMode(motorIn2, OUTPUT);
```

```
pinMode(greenLED, OUTPUT);
}

void loop()
{
    int sensorValue = analogRead(tempPin);

    float voltage = sensorValue * (5.0 / 1023.0);

    float celsius = (voltage - 0.5) * 100.0;

    Serial.print("Temperature: ");
    Serial.print(celsius);
    Serial.println(" C");

    digitalWrite(greenLED, HIGH);

    if (celsius < 25)
    {
        analogWrite(motorEnable, 0);
    }
    else if (celsius >= 25 && celsius <= 35)
    {
        int speed = map((int)celsius, 25, 35, 100, 255);

        digitalWrite(motorIn1, HIGH);
        digitalWrite(motorIn2, LOW);
    }
}
```

```
    analogWrite(motorEnable, speed);
}
else
{
    digitalWrite(motorIn1, HIGH);
    digitalWrite(motorIn2, LOW);

    analogWrite(motorEnable, 255);
}

delay(500);
}
```

Working Principle

1. The TMP36 temperature sensor continuously measures the ambient temperature.
2. The Arduino reads the analog voltage from the sensor through pin A0.
3. The sensor voltage is converted into temperature in degrees Celsius using the TMP36 conversion formula.
4. The measured temperature is displayed on the Serial Monitor in real time.
5. When the temperature is below 25°C, the fan remains OFF.
6. When the temperature is between 25°C and 35°C, the fan speed gradually increases using PWM control.
7. When the temperature exceeds 35°C, the fan runs at maximum speed.
8. The green LED remains ON to indicate that the system is powered and operational.

Results

The system successfully monitored ambient temperature and automatically controlled the fan speed based on temperature conditions. Real-time temperature values were displayed on the Serial Monitor, and PWM control enabled smooth variation of fan speed. The LED indicator provided visual confirmation that the system was operating correctly.

Conclusion

This project demonstrates the integration of sensors, motor control, and Arduino programming to create an intelligent temperature-responsive cooling system. The use of PWM allows efficient fan control, making the system suitable for automation and embedded system applications.